

Urban Users' Perceptions of UPI Usage: A Case Study of Chennai

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Abstract:

The National Payments Corporation of India created the Unified Payments Interface (UPI), which has revolutionised digital payments by allowing for real-time interbank, peer-to-peer (P2P), and peer-to-merchant transactions. This study looks into the engagement habits of urban UPI users in Chennai, with an emphasis on two different demographic groups: students and working professionals. It investigates why people choose QR code scanning over other UPI methods, focussing on the importance of simplicity, speed, and perceived security in determining user behaviour. The study used a cross-sectional survey approach, with 102 respondents completing structured questionnaires. The key findings show no statistically significant difference in the frequency of UPI usage between students and working professionals. However, QR code scanning has emerged as the favoured approach due to its ease, quickness, and improved security perceptions. This study addresses the gap in understanding demographic-specific usage trends and feature-specific preferences, providing stakeholders with relevant data. These findings can help to optimise UPI features to better serve varied urban demographics, hence contributing to the growth of inclusive and efficient digital payment ecosystems.

INTRODUCTION:

In this Digital era, the Digital payments have transformed transaction methods worldwide. In India, the Unified Payments Interface (UPI) has become a game-changer in this space. Introduced in 2016 by the National Payments Corporation of India (NPCI), UPI enables instant and seamless fund transfers, fostering seamless peer-to-peer (P2P) and peer-to-merchant (P2M) transactions, offering a highly efficient alternative to conventional payment modes such as cash, debit/credit cards, and net banking. It is a real-time payment system which enables instant fund transfers between bank accounts via a mobile application using a UPI ID. Users can generate a UPI ID through a UPI enabled app by linking it to their mobile number and setting a secure PIN. Transactions can be initiated using methods such as QR code scanning, UPI IDs, or mobile numbers, offering a cashless payment experience. UPI can function on a smartphone, acting like an email ID for money transactions (Mohapatra, 2017). UPI was developed using a mix of programming languages, including Java, Python, and C++, and incorporates secure encryption protocols to ensure transaction safety and integrity (Bilal & K, 2024). With over 10 billion transactions processed in October 2023 alone, UPI's widespread adoption has positioned

India as a global frontrunner in digital payments. By linking digital payment objectives with financial inclusion, the UPI system exemplifies its commitment to bridging the gap between technology and accessibility, providing easy access to digital payments, reducing reliance on cash, and promoting inclusivity in financial systems, which is a core goal of governance in a digital economy. UPI provided relief during the demonetization of 1,000 and 500-rupee notes by the Indian government in 2016. Initiatives like Digital India have laid the groundwork for UPI's success, with over 500 million bank accounts opened (Pradhan Mantri Jan Dhan Yojana., 2023). The Covid-19 pandemic also played a crucial role in promoting digital payments in India. Compared to semi-urban and rural areas, urban areas, being hubs of commerce and connectivity, have witnessed significant UPI adoption due to the convenience it offers. QR code scanning, in particular, has become a preferred method for digital payments in cities, enabling users to complete transactions quickly and securely without the need for physical cash or cards. However, preferences and usage patterns among different demographic groups, such as students and working professionals, remain underexplored. Understanding these patterns can provide valuable insights for policymakers, businesses, and app developers to enhance the digital payment ecosystem. Therefore, the paper examines the following research questions:

RQ 1: Do students in Chennai use UPI more frequently than working professionals for daily transactions such as transportation, food, and shopping?

RQ 2: To what extent do UPI users in Chennai prefer QR code scanning over other UPI methods (e.g., UPI ID entry) due to factors like convenience, speed, and perceived security?

This study focuses on Chennai, a metropolitan city with a growing digital economy, to examine urban users' perceptions of UPI compared to traditional payment methods. It specifically investigates whether students in Chennai use UPI more frequently than working professionals and explores the preference for QR code scanning over other UPI methods like entering a UPI ID to transfer money. The findings aim to deepen the understanding of user behavior and satisfaction in the context of digital public governance. By addressing these questions, the study seeks to bridge the gap between technological innovation and user-centric design in digital payment systems. The insights gained can help stakeholders identify opportunities to optimize UPI for diverse urban demographics, ensuring the digital payment landscape continues to evolve inclusively and effectively.

THEORETICAL FRAMEWORK:

In this study, the Unified Theory of Acceptance and Use of Technology (UTAUT), developed by Venkatesh et al. (2003), offers a comprehensive framework for understanding technology adoption and use. By synthesizing elements from eight established models of technology acceptance, UTAUT provides an integrated perspective that captures the multifaceted factors influencing user behavior. The framework is particularly suited for this study, as it addresses the interplay of performance, effort, social, and infrastructural factors that drive the adoption of digital payment systems like the Unified Payments Interface (UPI) in urban India.

UTAUT is built around four core constructs that collectively explain users' intentions and behavior regarding technology adoption:

1. **Performance Expectancy (PE):** PE is the extent to which a person believes that utilising a specific technology would improve their performance or help them achieve their objectives. This construct assesses how customers view UPI's capacity to offer prompt, dependable, and effective payment solutions in contrast to more conventional approaches. Adoption is more likely when performance expectations are high, especially in urban settings where speed and efficiency are valued (Rizkalla et al., 2024).
2. **Effort Expectancy (EE):** This term refers to how easy a technology is thought to be to operate. User-friendly and intuitive technologies have a higher chance of being accepted, particularly by people with different degrees of technological expertise (Ayaz & Yanartaş, 2020). UPI is a desirable choice for a variety of demographic groups because of its straightforward features, such as QR code scanning and smooth app interfaces, which reduce the amount of work needed to conduct transactions.
3. **Social Influence (SI):** SI measures how much people think that significant others think they ought to use a particular technology (Joa & Magsamen-Conrad, 2022). Social norms, peer recommendations, and cultural trends are frequently the driving forces behind the quick uptake of digital payment systems like UPI in metropolitan settings. When friends, family, or coworkers strongly support UPI, users may be persuaded to use it as well, particularly in an urban ecosystem that is connected and tech-savvy.
4. **Facilitating Conditions (FC):** These include the infrastructure and resources that enable the usage of technology, such as device access, dependable internet connectivity, and the presence of UPI-accepting merchants. Strong digital infrastructure increases adoption rates in India's cities. Effective use, however, can be hampered by issues like

erratic connectivity or technological obstacles, underscoring the significance of favourable circumstances for long-term adoption (Peñarroja et al., 2019).

By applying the UTAUT framework, this study aims to explore the interplay of these constructs in shaping UPI adoption among students and working professionals in Chennai. UTAUT's ability to integrate individual, social, and contextual factors makes it a valuable tool for understanding the diverse influences on user behavior in urban digital payment ecosystems.

LITERATURE REVIEW:

UPI Adoption:

UPI's user-friendly design, interoperability, and real-time transaction capabilities have made it a cornerstone of India's cashless economy. The rapid adoption of UPI, particularly in urban areas, highlights its potential to transform financial behavior and support digital public governance initiatives. Theoretical frameworks, including the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), and Diffusion of Innovation (DOI) theory, have been widely applied to understand UPI's adoption and its impact on user behavior. This review synthesizes insights from these frameworks, focusing on key factors influencing UPI adoption, user satisfaction, and challenges in urban contexts like Chennai.

Theoretical Frameworks for UPI Adoption:

1. **Technology Acceptance Model (TAM):** The TAM framework has been instrumental in analyzing UPI adoption. S, (2019) applied TAM in Mysuru to evaluate Perceived Usefulness (PU), Perceived Ease of Use (PEOU), and Behavioral Intention (BI) to adopt UPI. The study found that users prioritize the speed and convenience offered by UPI, which reduces reliance on traditional banking methods. Ease of navigation and minimal technical complexity were also critical in driving adoption.
2. **Unified Theory of Acceptance and Use of Technology (UTAUT):** Razi-ur-Rahim et al, (2024) employed UTAUT to explore Performance Expectancy, Effort Expectancy, and Social Influence in the context of UPI adoption in India. Their findings emphasize that users adopt UPI primarily because they perceive it as efficient and reliable. Social factors, such as recommendations from peers, significantly influence user behavior, underscoring the role of observability in adoption trends.

3. **Diffusion of Innovation (DOI) Theory:** The DOI theory provides a broader perspective by focusing on the attributes of innovation that drive adoption (Fahad & Shahid, 2022). As applied to UPI, the theory highlights five critical factors:

- **Relative Advantage:** UPI's benefits, such as speed, cost-effectiveness, and convenience, position it as a superior alternative to cash and card payments.
- **Compatibility:** UPI's seamless integration with smartphones and banking systems aligns with users' existing financial practices and habits.
- **Complexity:** A user-friendly interface is essential to overcoming the perceived difficulty of adopting digital payment platforms.
- **Observability:** Seeing peers use and endorse UPI builds trust and encourages adoption.
- **Trialability:** Features like easy onboarding and trial transactions reduce hesitation, facilitating a smoother adoption process.

User Preferences and Satisfaction:

User satisfaction is a central theme in UPI adoption studies. M. N. Prakash, (2023) examined UPI usage among university students in Madikeri City and found that awareness, preference, and satisfaction levels were significant in this demographic. Students favored UPI for its convenience in daily transactions, such as paying for food and transportation. The study recommended increasing awareness to boost adoption further, particularly among less informed groups. The preference for QR code scanning as a UPI method has been widely noted. Sousa & Cuncoienar, (2023) reported that QR codes are valued for their simplicity and speed, eliminating the need for manual entry of payment details. This feature resonates with urban users, who often prioritize efficiency in financial transactions. However, challenges such as transaction failures and connectivity issues persist, highlighting areas for improvement to enhance user experience.

Despite its advantages, UPI faces several challenges that limit its potential:

1. **Technical Barriers:** Transaction failures, poor internet connectivity, and delayed settlements undermine user trust. Unlike systems like Kenya's M-Pesa, which rely on SMS-based technology, UPI's dependence on internet access poses a barrier in areas with inconsistent connectivity.

2. **Security Concerns:** Users remain apprehensive about fraud and data breaches, even with robust authentication protocols. Addressing these concerns is critical to sustaining UPI's growth.
3. **Infrastructure Demands:** The scalability of UPI to handle a growing user base requires continuous infrastructural upgrades, especially as adoption expands beyond urban centers.

Research Gaps:

Existing studies provide valuable insights into UPI adoption and user behavior, but significant gaps remain:

1. **Urban-Specific Research:** Most studies analyze general user behavior, with limited focus on metropolitan areas like Chennai, where urban dynamics significantly influence digital payment adoption.
2. **Demographic-Specific Trends:** Variations between groups such as students and working professionals are underexplored, hindering the development of targeted strategies for adoption.
3. **Feature-Specific Preferences:** While QR codes are recognized as a preferred UPI method, limited research investigates user preferences compared to other options like UPI IDs or phone numbers.

UPI, with its features of instant transfers and broad accessibility, has revolutionized payments in urban India. However, technical barriers, such as inconsistent internet connectivity, data privacy concerns, and infrastructural demands for QR code scanning, must be addressed to sustain growth and ensure inclusivity. By bridging these gaps, this study aims to contribute to the discourse on financial inclusion and digital governance.

RESEARCH DESIGN:

In order to investigate UPI usage patterns and preferences among urban Chennai residents, this study uses a cross-sectional survey design. To make sure the sample population was representative and diverse, a stratified random sampling technique was used. There were two main demographic strata into which the respondents were separated:

- Students (18–24 years): Pursuing higher education.
- Working Professionals (25–60 years): Employed full- or part-time.

Random sampling within these strata ensured a balanced representation of age, frequency of UPI usage, and occupation. This method was chosen to capture demographic-specific behaviors effectively, as students and professionals are significant contributors to UPI adoption in urban areas. This design identifies patterns and factors influencing QR code preferences within the constraints of time and resources.

DATA COLLECTION:

An online survey aimed at Chennai UPI users served as the main instrument for gathering data. The Unified Theory of Acceptance and Use of Technology (UTAUT) framework, which looks at four constructs—Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions—was the basis for the survey's design. In order to preserve anonymity, respondents' personal information was anonymised and their participation was voluntary.

Survey Design:

The questionnaire was divided into three distinct sections:

1. **General Information:** Important demographic information (age group, occupation) and baseline data (frequency of UPI usage) were recorded in this area. In order to offer context for examining usage patterns, it created a demographic profile.
2. **UPI Usage Preferences and User Perceptions:** This section evaluated the acceptance of UPI techniques such as entering a phone number, scanning a QR code, and entering a UPI ID. Preferences for QR codes were examined in this part, taking into account elements like convenience, security, speed, and ease of use. Additionally, it employed a 5-point Likert scale to gauge opinions regarding the efficiency and time-saving advantages of QR code scanning.
3. **Difficulties and Suggestions:** This section listed the difficulties encountered when utilising QR codes, such as problems with internet access, unsuccessful scanner detection attempts, or transaction errors. Additionally, open-ended questions were incorporated to collect practical recommendations for enhancing UPI systems. Perceptions of UPI's general efficacy in enabling smooth transactions were also investigated.

UTAUT Constructs and Measurement

1. **Performance Expectancy (PE):** Measured by questions about perceived speed, efficiency, and security of UPI transactions.

2. **Effort Expectancy (EE):** Determined by how simple QR codes are thought to be to use in comparison to other approaches.
3. **Social Influence (SI):** Assessed by enquiries on how peers or coworkers affect the adoption of UPI.
4. **Facilitating Conditions (FC):** Examined using open-ended enquiries on more general problems and user recommendations for enhancements, as well as structured queries concerning technical difficulties.

Sample Size and Confidence Level:

Data was collected from 102 respondents, with 50 students, 50 working professionals and 2 unemployed people. The margin of error for this sample size, assuming a 95% confidence level and a population proportion of 50%, is approximately 9.68%. This provides a reasonable level of confidence in the findings.

Data Analysis:

The Mann-Whitney U test was used to compare the frequency of UPI usage between working professionals and students. When comparing two independent groups and the dependent variable is ordinal and normality is not expected, this non-parametric test works well (Najab and McKnight, 2010). With the frequency of UPI usage (categorised as daily, 2-3 times per week, or rarely) as the dependent variable and occupation (categorised as students or working professionals) as the independent variable, the study looked at the association between occupation and UPI usage frequency. The findings imply that the frequency of UPI usage in Chennai does not differ statistically significantly between working professionals and students. The lack of a significant difference in UPI usage frequency between students and working professionals highlights the widespread adoption of UPI across demographics. Factors such as accessibility and ease of use may have contributed to this parity.

The study also aimed to identify factors influencing the preference for QR code scanning, where the dependent variable was the preference for QR code scanning and the independent variables were convenience, speed, and perceived security (ordinal). Ordinal logistic regression was then performed to assess the impact of these factors on the preference for QR code scanning. Ordinal logistic regression is suitable for analyzing the relationship between an ordinal dependent variable and one or more independent variables. It is ideal for identifying which factors (convenience, speed, perceived security) most influence QR code preference. The results, indicated that all three factors like convenience, speed, and perceived significantly influence the preference for QR code scanning. The preference for QR code scanning among UPI users

underscores the importance of convenience, speed, and security. These findings align with existing literature on digital payment adoption, emphasizing the role of user experience in technology acceptance. UPI has the potential to enhance multiple sectors, such as mobile payments, high-value transactions, and even cross-border payments, by bringing them to an unparalleled level of efficiency and convenience (Khan, 2023).

Key Takeaways

1. **RQ1:** There is no significant difference in the frequency of UPI usage between students and working professionals in Chennai.
2. **RQ2:** Convenience, speed, and perceived security significantly influence the preference for QR code scanning among UPI users in Chennai.

Limitations of the Study:

While this study offers valuable insights into UPI usage patterns and preferences among urban users in Chennai, several limitations should be noted. The sample size of 102 participants is relatively small compared to the city's large population, which may limit the diversity and variability of the behaviors captured. Additionally, the use of a non-probability sampling method could introduce sampling bias, potentially leading to the underrepresentation or overrepresentation of certain demographic groups, thus affecting the generalizability of the findings. Self-reported data also present challenges, including potential biases such as social desirability, recall inaccuracies, and response inaccuracies. Geographically, the focus on urban Chennai restricts the applicability of the findings to rural or semi-urban areas, where infrastructure, technology access, and socioeconomic factors differ significantly. Furthermore, the cross-sectional survey design captures data at a single point in time, limiting the study's ability to assess trends or causal relationships. Lastly, while the study examines factors like convenience, speed, and perceived security, it does not consider other elements, such as user experience with specific apps, peer influence, and promotional offers.

Conclusion:

Despite these limitations, the study offers useful insights of UPI usage patterns and preferences among urban users in Chennai. The findings enhance our understanding of digital payment adoption in urban India, emphasizing the significance of factors like convenience, speed, and perceived security in shaping user preferences for QR code scanning. Future research with larger and more diverse samples, longitudinal designs, and the inclusion of additional factors could yield a more comprehensive understanding of UPI usage behaviors and preferences.

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Table: Frequency of UPI Usage by Occupation

Occupation	Daily	2-3 times a week	Rarely	Total
Students	25	12	10	50
Working Professionals	28	15	10	50
Total	53	27	20	100

Mann-Whitney U Test Results:

Group Comparison	U- Statistic	p-value
Students vs Working Professionals	1225.5	0.45

U-statistic value is the test statistic that corresponds to the amount of overlap between the two groups' rankings.

Interpretation: If $p < 0.05$, there is a statistically significant difference between the two groups.

Table 2

Ordered Logistic Regression Results for preference of UPI QR Code Scanning

Predictor variables	Coefficient (B)	Std. Error (SE)	z-value (z)	p-value (p)	95% Confidence Interval [Lower, Upper]
Intercept	0.4567	0.234	1.953	0.051	[-0.002, 0.915]
Convenience	0.6789	0.312	2.175	0.030	[0.067, 1.291]
Speed	0.5432	0.278	1.953	0.051	[-0.002, 1.088]
Security	0.6123	0.295	2.075	0.038	[0.033, 1.191]
Threshold	1.2345	0.456	2.707	0.007	[0.340, 2.129]

Interpretation:

1. **Intercept:** The baseline level of preference for QR code scanning.
2. **Convenience Rating:** A positive coefficient (0.6789) indicates that higher convenience ratings are associated with a higher likelihood of preferring QR code scanning. The p-value (0.030) is less than 0.05, indicating statistical significance.
3. **Speed Rating:** A positive coefficient (0.5432) indicates that higher speed ratings are associated with a higher likelihood of preferring QR code scanning. The p-value (0.051) is slightly above 0.05, indicating marginal significance.

4. **Security Rating:** A positive coefficient (0.6123) indicates that higher security ratings are associated with a higher likelihood of preferring QR code scanning. The p-value (0.038) is less than 0.05, indicating statistical significance.
5. **Threshold:** The threshold value (1.2345) represents the cut-off point on the latent variable scale that separates different levels of the ordinal outcome.